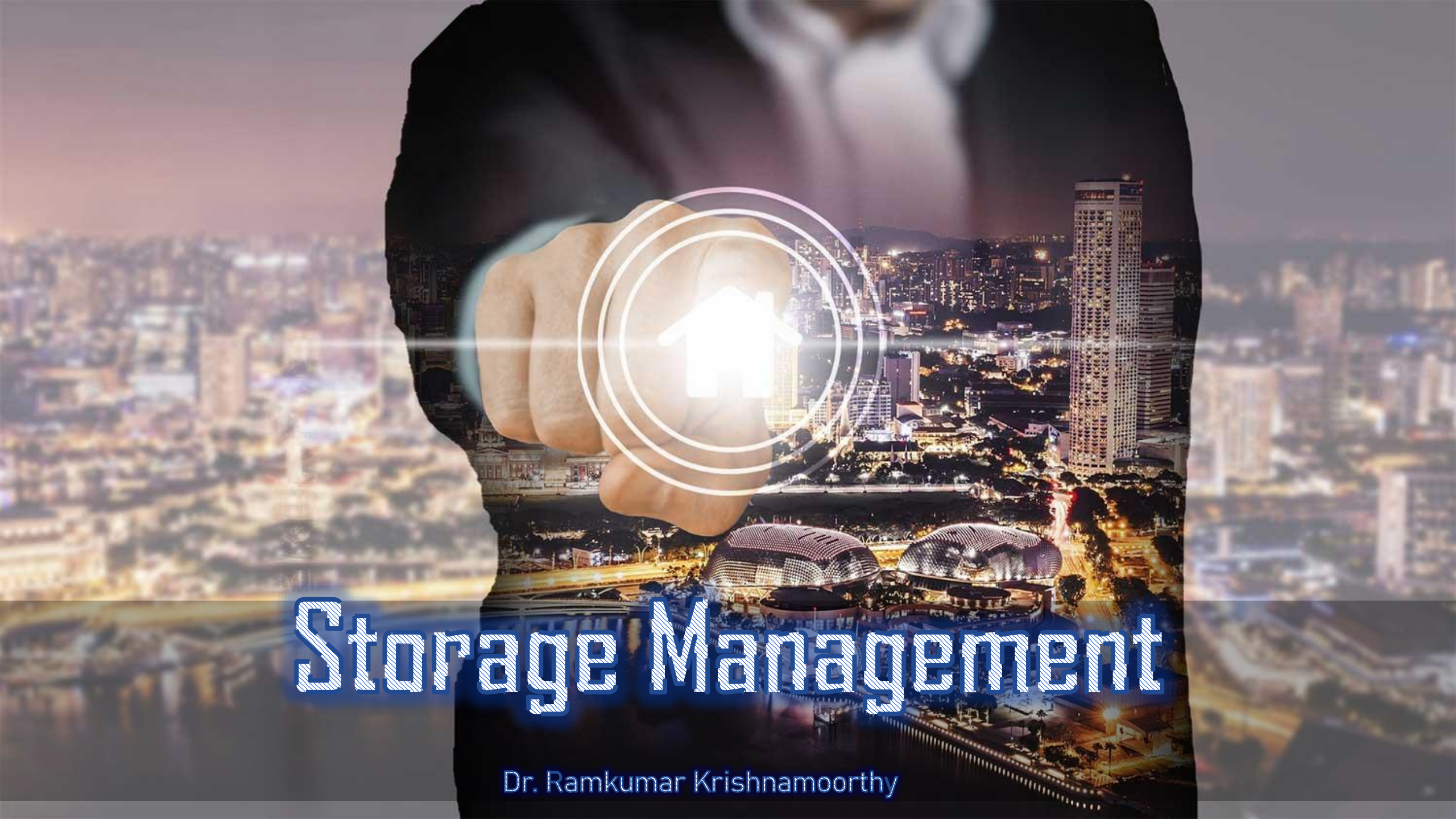


A hand in a dark suit jacket points towards a glowing white house icon. The icon is surrounded by three concentric white circles. The background is a blurred night cityscape with lights from buildings and streets. The overall image has a dark, moody aesthetic with a focus on the hand and the house icon.

Storage Management

Dr. Ramkumar Krishnamoorthy

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Information Storage

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Data vs Information



NEWS

Rohit to lead India at 2024 T20 World Cup, Dravid to remain head coach

BCCI secretary Jay Shah confirms as much; also says it's mandatory for centrally contracted players to play domestic cricket when fit

PTI and ESPNcricinfo staff
14-Feb-2024



Watch: Rohit Sharma Sledges Dinesh Karthik With 'World Cup' Remark. Video Goes Viral

Rohit Sharma is well known for his hilarious comments on the field and the stump mic captured another gem from him during the IPL 2024 match between Mumbai Indians and Royal Challengers Bengaluru.

NDTV Sports Desk Updated: April 12, 2024 07:50 AM IST

Read Time: 4 min



Data vs Information



Infosys
@Infosys

“The inauguration of our Raleigh Technology and Innovation Hub empowers our NC employees by providing them with pioneering technology training that will help American industries revitalize their core businesses,” says Salil Parekh, CEO, Infosys. [#USJobs](#)



Data

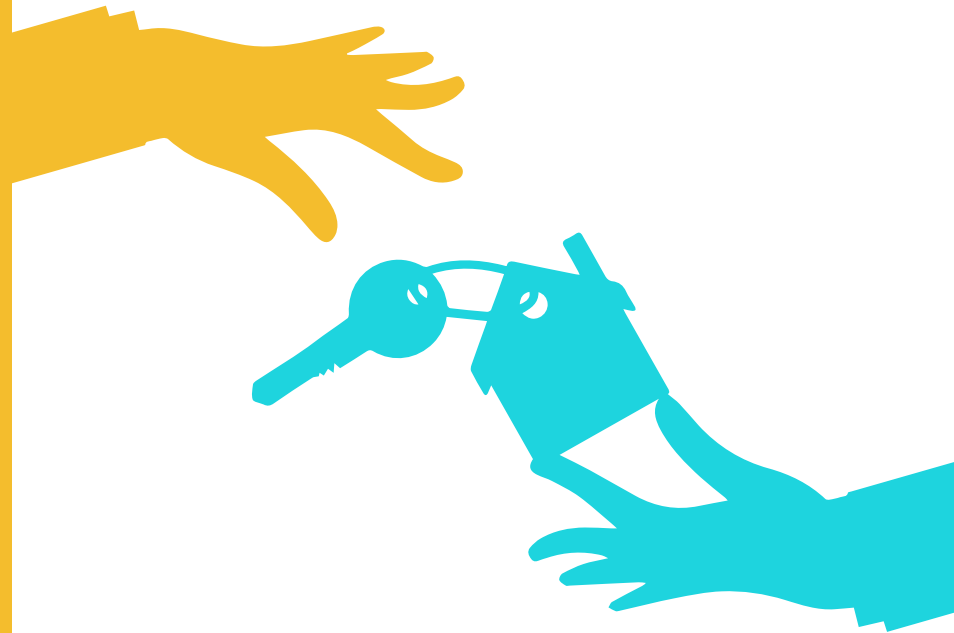
Definition

- It is an un processed Information
- Data is a collection of raw facts from which conclusions might be drawn

“

Data vs. Information

”



Related

Information

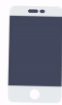
Definition

- It is a processed Data

Data vs Information

Data	Information
Data is unorganised and unrefined facts	Information comprises processed, organised data presented in a meaningful context
Data is an individual unit that contains raw materials which do not carry any specific meaning.	Information is a group of data that collectively carries a logical meaning.
Data doesn't depend on information.	Information depends on data.
Raw data alone is insufficient for decision making	Information is sufficient for decision making
An example of data is a student's test score	The average score of a class is the information derived from the given data.

Data vs Information - Case Study





Information Storage

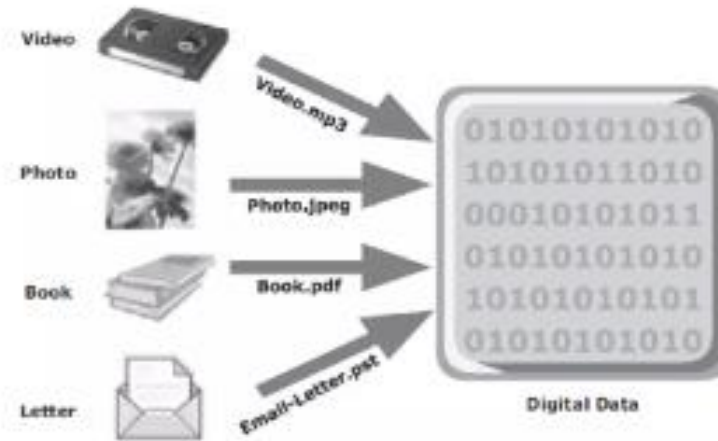
- **Information Storage – Central pillar for IT**
 - Traditional (Data stored in HDD, CD, Floppy, etc.)
 - Now?
 - Flash Drives, External Storage, On the cloud?
- **Grown into Highly sophisticated technology**
 - Provides solutions for creating, storing, managing, connecting, protecting, securing, sharing & optimizing digital information

Information Storage

- **Need of Information Storage Professionals/Technologists**
 - Today's even seasoned IT professionals (including application, systems, database, and network administrators) do not share a common foundation about how storage technology affects their areas of expertise.
 - Increasing demand for personnel who understand data storage and the value of data to businesses.
- **It would be beneficial to examine**
 - Who is creating data?
 - What types of data are being created?
 - When data becomes information?

Data & Information (Sharing)

- **Digital Data**



- **Information is required On-Command & On-Demand**
 - Eg. Over Internet Search, Email, Social Network
- **Data Sharing**
 - Upload into Data Centers via Internet
 - Virtuous Cycle of Information

Information Storage – Data Creation

- **Information is a need of daily life.**
- **Increasing Rate of Data Creation => Increasing need of Data Storage**
 - Data is generated in excess of 50% year-over-year
 - Store data over longer periods of time?
 - Store data forever?
 - Accessibility?
- **IT budget should keep up with data storage needs**
 - Includes expenditure on Servers, Networks, Storage, Personnel, etc.
 - IT expenditure on Storage has increased proportionally - about 40%

Information Storage – Data Creation

- **Individuals**
 - **More Data created by Individuals than at business**
 - **What data is created?**
 - Photos, Documents, Spreadsheets, Video, etc.
 - **Where is the data stored - locally?**
 - Cameras, MP3 players, Laptop HDD, CDROM/DVDs, USB, etc.
 - Challenge !!
 - Managing data stored in diverse location

Information Storage – Data Creation

- **Business**
 - **What data is created?**
 - The data that a business collects is about their customers, partners, products and services.
 - Product data: inventory, description, pricing, availability, sales numbers and projections
 - Customer data: orders, shipping details
 - Account data: banking, financial services industry
 - Medical data: health care providers, insurance industry, hospitals

Information Storage – Data Creation

- **Business**
 - **Where is Data stored?**
 - **Depends on size of Business**
 - Local, individual work stations or on centralized disk array systems, Servers, Tapes, CDROM/DVDs, Off-site libraries
 - **Challenge !!**
 - Information is stored securely and accurately
 - Information should be available “on demand” 24/7

Data Creation – Case Study Exercise



Information Storage – Value of Data to Business

- **What do businesses “do” with the data they collect?**
 - **Businesses “mine” the “Data” and turn it into “Information”**
 - Extract meaningful patterns or trends
 - Businesses create information to better manage the business.
- **Examples of information include**
 - **Buying habits and patterns of customers**
 - **GPS locations of delivery trucks**
 - **Health history of patients**
 - **Locations where a credit card is used**

Information Storage – Value of Individual Data to Business

- **What data created by individuals might be valuable to a business?**
- **Examples of business value from individuals' data include:**
 - On-line resume storage and management service
 - On-line photo storage and organizer
 - **Discussion: Job Search Engine**

Information Storage – Value of Information to Business

- **Identifying new business opportunities**
 - **Buying/spending patterns:** Internet stores, retail stores, supermarkets
 - **Customer satisfaction/service:** tracking shipments & deliveries
- **Identifying patterns that lead to changes in existing business. For example:**
 - **Reduced cost:** delivery service optimizing utilization of vehicles and gas
 - **New products:** MP3 player speaker systems
 - **New services:** security alerts for “stolen” credit card purchases
 - **Targeted marketing campaigns:** communicate to bank customers with high checking account balances about a special savings plan
- **Creating a competitive advantage!**
- **Discussion: Bank – Special Savings Plan**

Information Storage – Information Availability

- **Information Availability** is defined as the ability to make information and related resources accessible as needed, when they are needed, where they are needed.

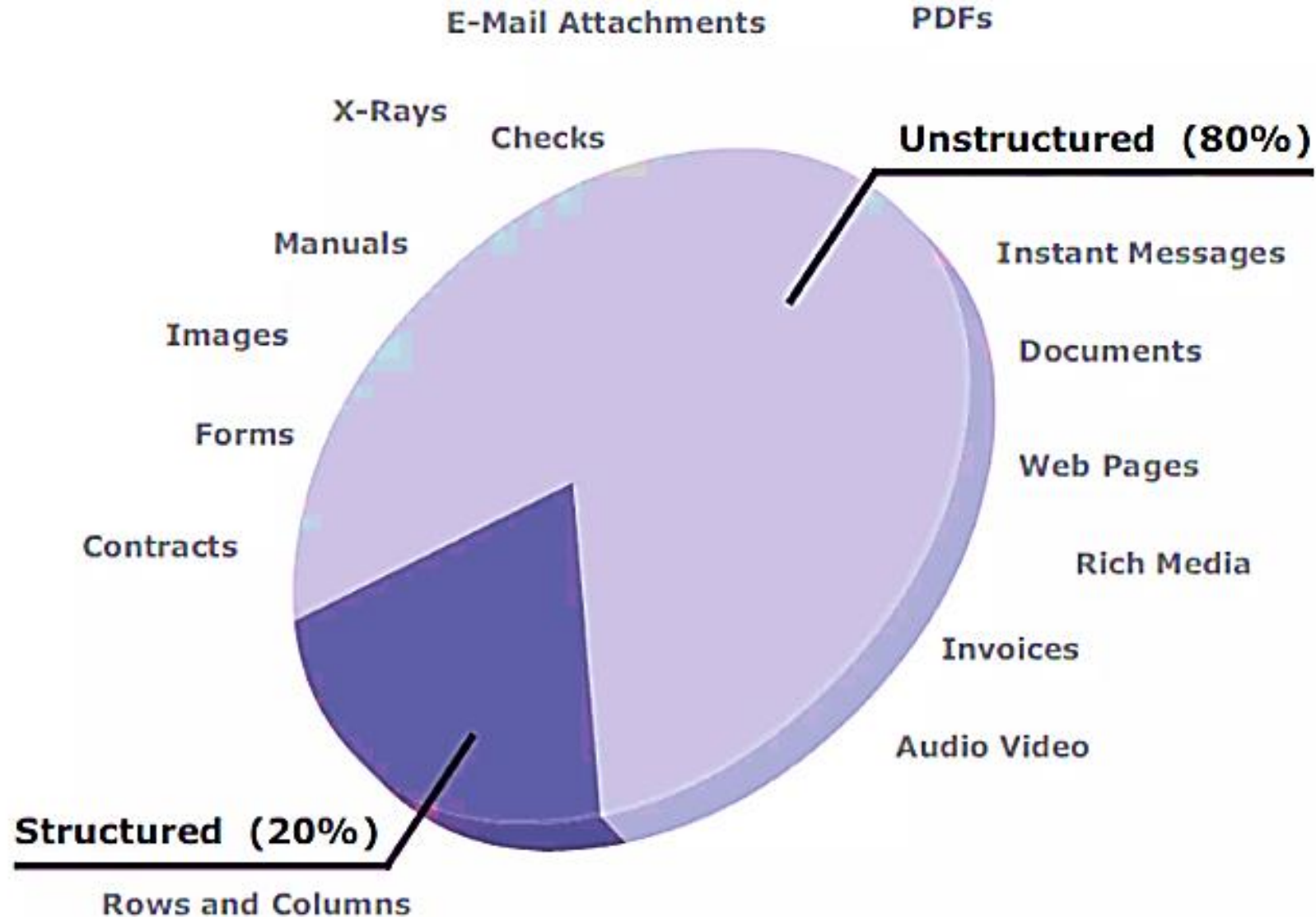
Information availability can be defined with the help of reliability, accessibility and timeliness

- **Reliability:** This reflects a component's ability to function without failure, under stated conditions, for a specified Amount of time.
- **Accessibility:** This is the state within which the required information is accessible at the right place, to the right user.
- **Timeliness:** Defines the exact moment or the time window (a particular time of the day, week, month, and/or year as specified) during which information must be accessible. For example, if online access to an application is required between 8:00 am and 10:00 pm each day, any disruptions to data availability outside of this time slot are not considered to affect timeliness.

Information Storage – Types of Data

- Data can be classified as structured or unstructured based on how it is stored and managed. Structured data is organized in rows and columns in a rigidly defined format so that applications can retrieve and process it efficiently.
- Structured data is typically stored using a database management system (DBMS).
- Data is unstructured if its elements cannot be stored in rows and columns, which makes it difficult to query and retrieve by applications.
- For example, customer contacts that are stored in various forms such as sticky notes, e-mail messages, business cards, or even digital format files, such as .doc, .txt, and .pdf.
- Due to its unstructured nature, it is difficult to retrieve this data using a traditional customer relationship management application.

Information Storage – Types of Data



Information Storage – Types of Data

- **Structured Data**
 - Formal & well organized
 - Stored in: Relational database or spreadsheet
- **Unstructured Data**
 - Informal, possibly text (such as XML tagged content) & disorganized
 - Stored in: Files as whole documents (or) in content management systems
 - Businesses generate unstructured data in the form of emails, forms, marketing materials, web pages, etc.
 - Over 80% of enterprise information is unstructured (Fulcrum Research, 2004).
 - Might require a lot more effort in searching, retrieving, and presenting to the end-user, than structured data.

Evolution of Storage Technology and Architecture

- **Centralized**
 - Terminals connected to a Mainframe computer which had connectivity to internal/external storage devices (disks, tapes).
 - Processes had to be in place for data access.
 - Introduces considerable delay in new application development and deployment.
 - Access to data (such as a report request or for archived data) was predicated on business needs.
 - Computational power was deemed more important than the immediacy of access.

Evolution of Storage Technology and Architecture

- **Decentralized/Distributed**
 - With advances in networking, the **client-server model** became prevalent
 - Business units within an enterprise could have access to their *own servers and storage*
 - Applications no longer had to wait in one central queue for data access and execution
 - Lead to fragmentation of information. It also becomes difficult to enforce uniform processes and policies, as well as to manage these **islands of information**

Evolution of Storage Technology and Architecture

- **Networked Storage**
 - Best practice model used in IT where data is located centrally and kept on disk in a storage system.
 - Data can be stored centrally. Access to it is allowed for different applications.
 - Network Storage allows servers to access data.
 - Each department (Production/HR/Finance/..) had its collection of clients, servers, and storage.
 - **Benefits**
 - Connects many computers to a central place for data storage and retrieval
 - Data can be more easily managed, shared and protected
 - Data can be made highly available

Evolution of Storage Technology and Architecture

- **Networked Storage**

- **Redundant Array of Independent Disks (RAID)**

- Addresses cost, performance & availability requirements of data
 - It continues to evolve & used in all storage architectures

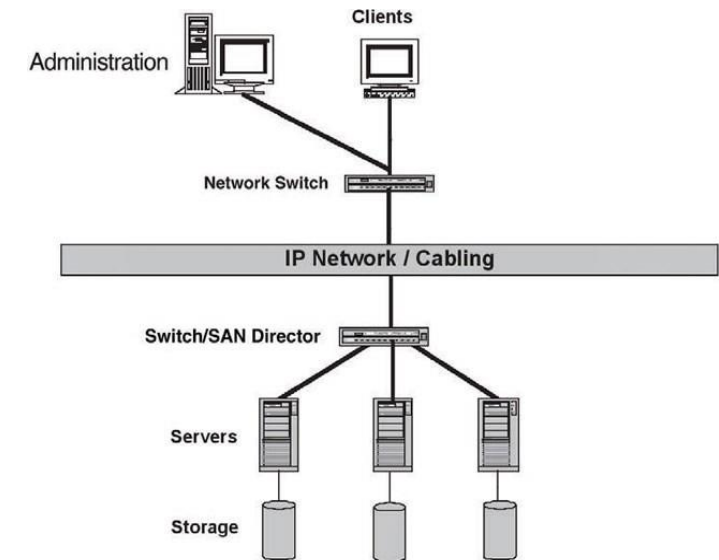
- **Direct Attached Storage (DAS)**

- **Connects directly** to a server or a group of servers (cluster)
 - Storage can be either **internal** (limited storage) or **external** to the server

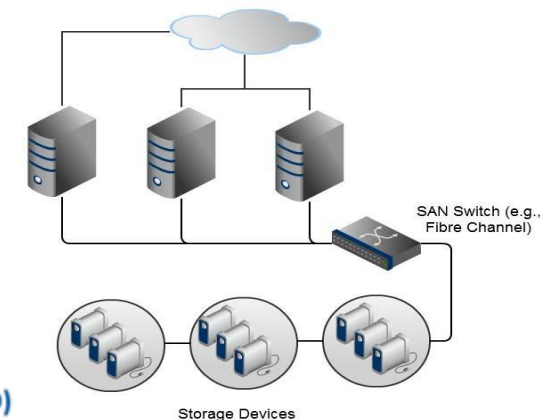
- **Storage Area Network (SAN)**

- Storage is **partitioned** & assigned to a server for data access
 - Dedicated, high-performance Fiber Channel (FC) network to facilitate **block-level communication** between servers & storage
 - Offers scalability, availability, performance & cost benefits compared to DAS

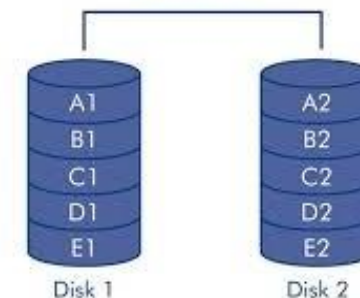
Direct Attached Storage (DAS)



Storage Area Network (SAN)



RAID 0



Redundant Array of Independent Disks (RAID)

Evolution of Storage Technology and Architecture

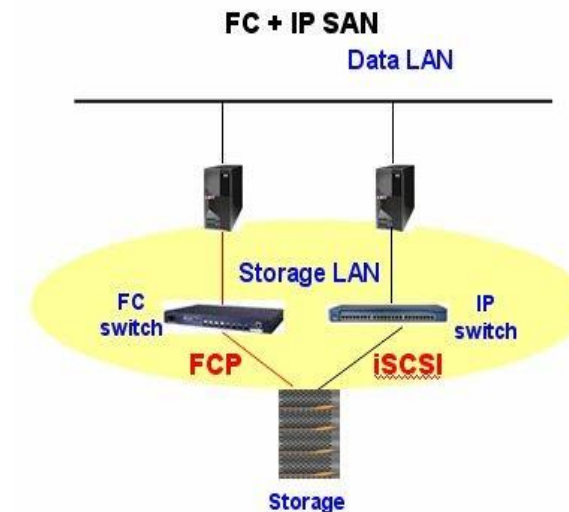
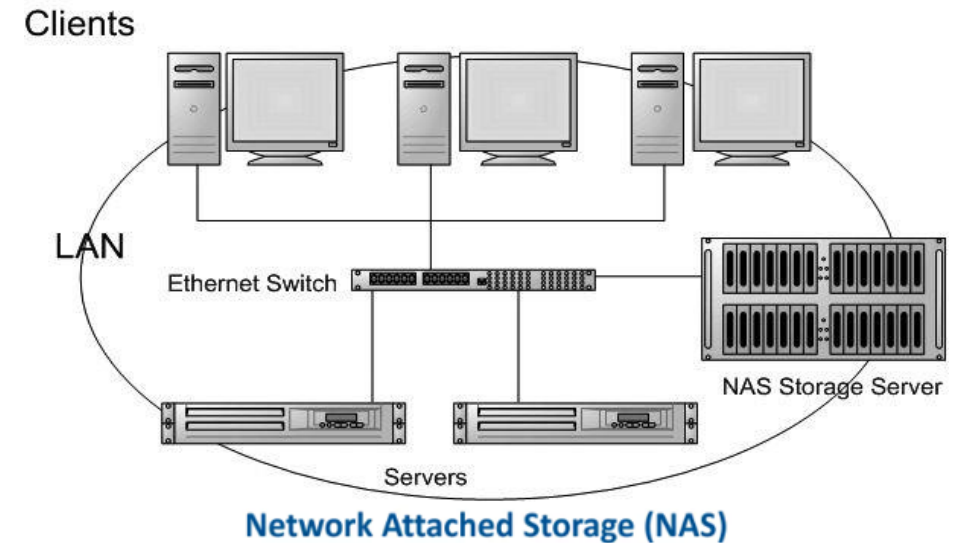
- **Networked Storage**

- **Network Attached Storage (NAS)**

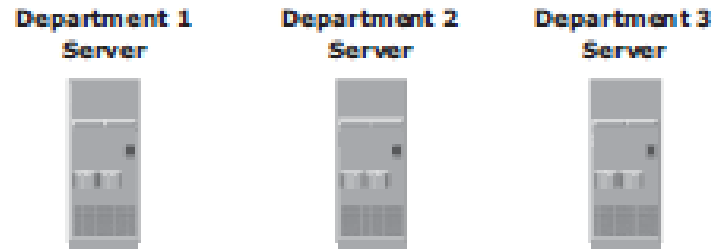
- **Dedicated storage** for file serving applications
 - Unlike a SAN, it **connects to an existing communication network (LAN)** and provides file access to heterogeneous clients
 - Offers higher scalability, availability, performance & cost benefits compared to general purpose file servers

- **IP-SAN**

- Convergence of technologies used in SAN & NAS
 - Provides **block-level communication** across LAN or WAN resulting in greater consolidation and availability of data

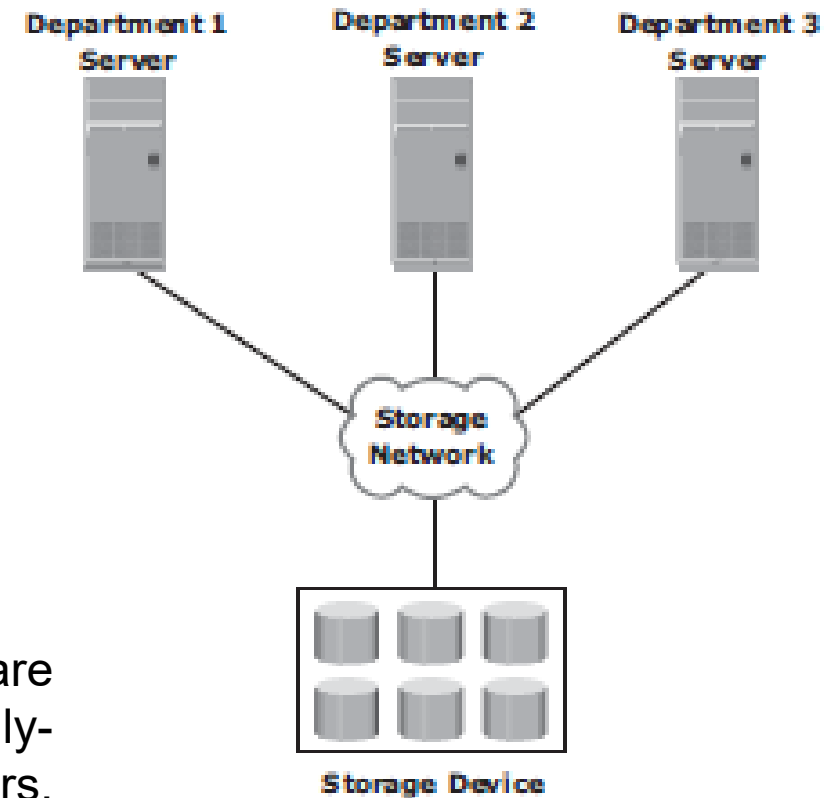


Evolution of Storage Technology and Architecture



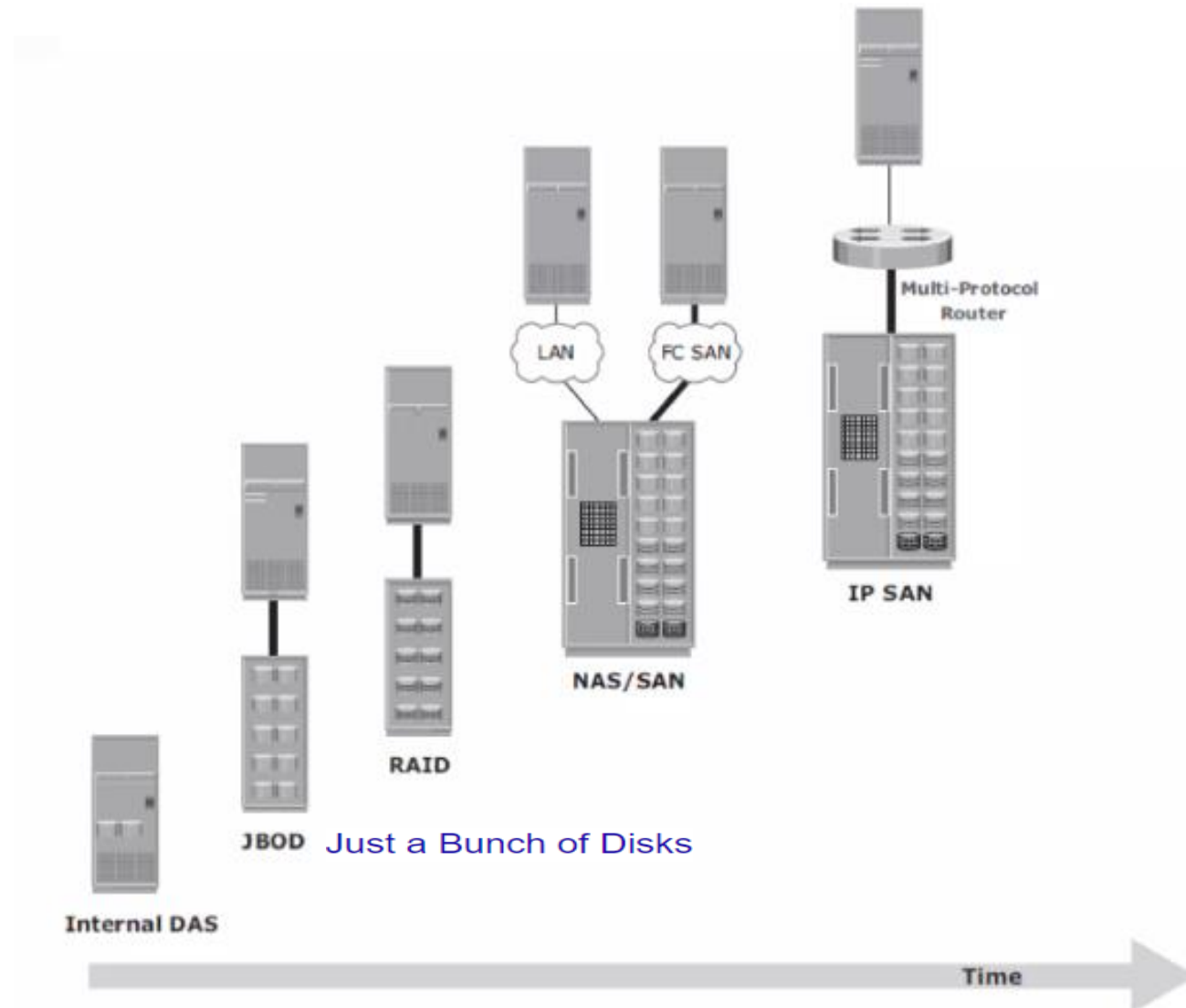
(a) Server-Centric Storage Architecture

- In earlier implementations of open systems, the storage was typically internal to the server.
- These storage devices could not be shared with any other servers. This approach is referred to as **server-centric storage** architecture
- In this **Information Centric architecture**, storage devices are managed centrally and independent of servers. These centrally-managed storage devices are shared with multiple servers. When a new server is deployed in the environment, storage is assigned from the same shared storage devices to that server.



(b) Information-Centric Storage Architecture

Evolution of Storage Technology and Architecture



Evolution of Storage Architectures

Data Center

- A data center is a physical facility that organizations use to house their critical applications and data.
- A data center's design is based on a network of computing and storage resources that enable the delivery of shared applications and data.
- The key components of a data center design include routers, switches, firewalls, storage systems, servers, and application-delivery controllers



Data Center – Components

- **Network infrastructure.** This connects servers (physical and virtualized), data center services, storage, and external connectivity to end-user locations.
- **Storage infrastructure.** Data is the fuel of the modern data center. Storage systems are used to hold this valuable commodity.
- **Computing resources.** Applications are the engines of a data center. These servers provide the processing, memory, local storage, and network connectivity that drive applications.

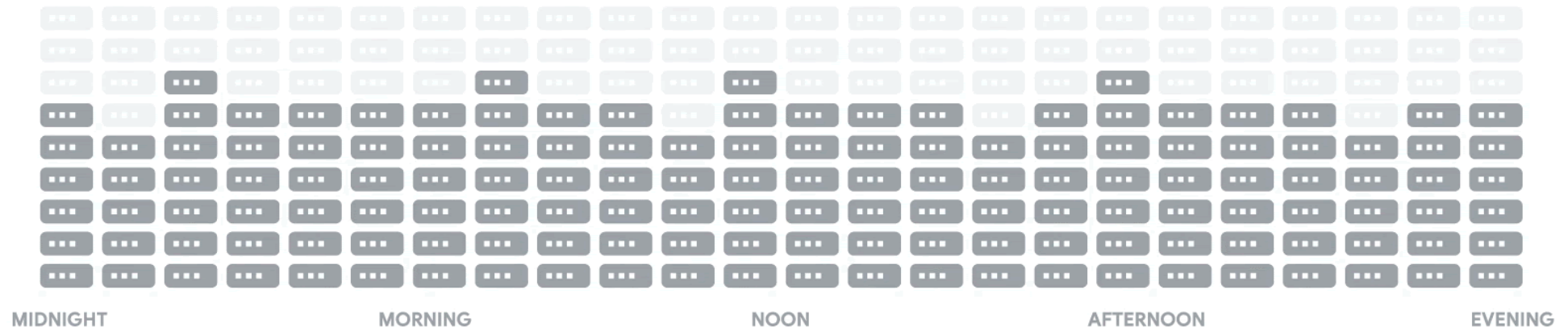
Data Center – Google



Data Center – Google Carbon-Intelligent Computing

Conventional compute load

Execution of compute tasks throughout the day, regardless of carbon impact



Visualization of how we shift compute tasks to different times of day to align with the availability of lower-carbon energy. In this illustration, wind energy in the evening and solar energy during the day

Data Center – Facebook



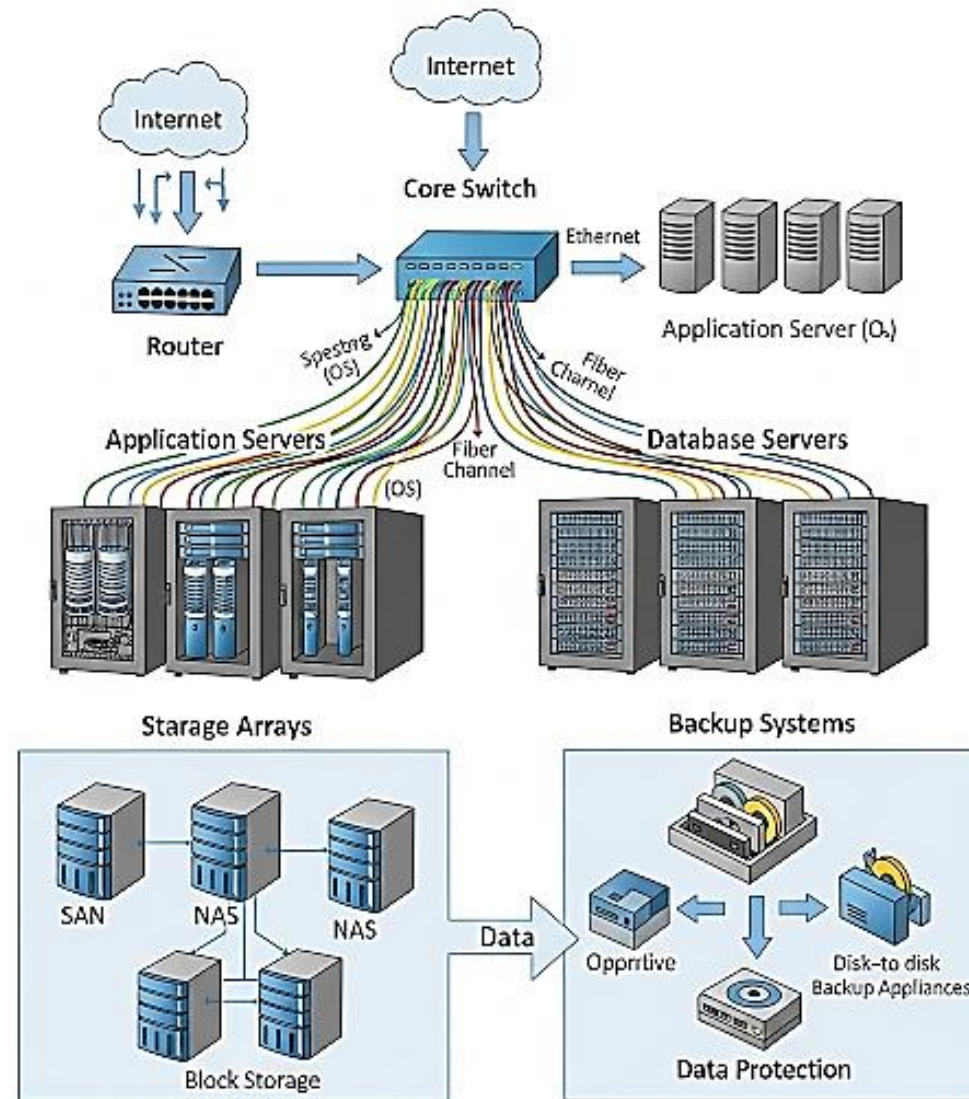
Data Center Infrastructure

- Organizations maintain data centers to provide centralized data-processing capabilities across the enterprise.
- Data centers house and manage large amounts of data.
- The data center infrastructure includes hardware components, such as computers, storage systems, network devices, and power backups; and software components, such as applications, operating systems, and management software.
- It also includes environmental controls, such as air conditioning, fire suppression, and ventilation.
- Large organizations often maintain more than one data center to distribute data processing workloads and provide backup if a disaster occurs.

Data Center Infrastructure – Core Elements

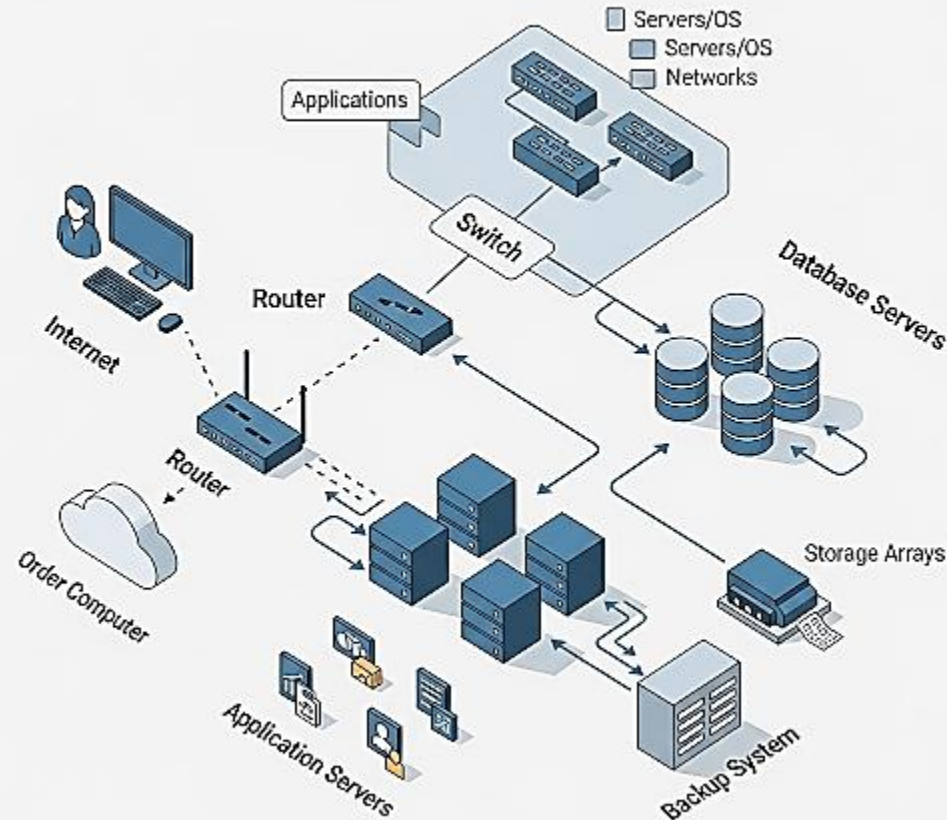
- **Applications**
 - Specialized and dedicated software to manipulate data
- **Databases**
 - DBMS & physical and logical storage of data
- **Servers/OS**
 - Provides computing platform required to run the applications and databases
- **Networks**
 - Provides data communication path between clients & servers or between servers and storage
- **Storage Arrays**
 - Place where “information lives”

Data Center Infrastructure – Core Elements



Data Center Infrastructure – Example

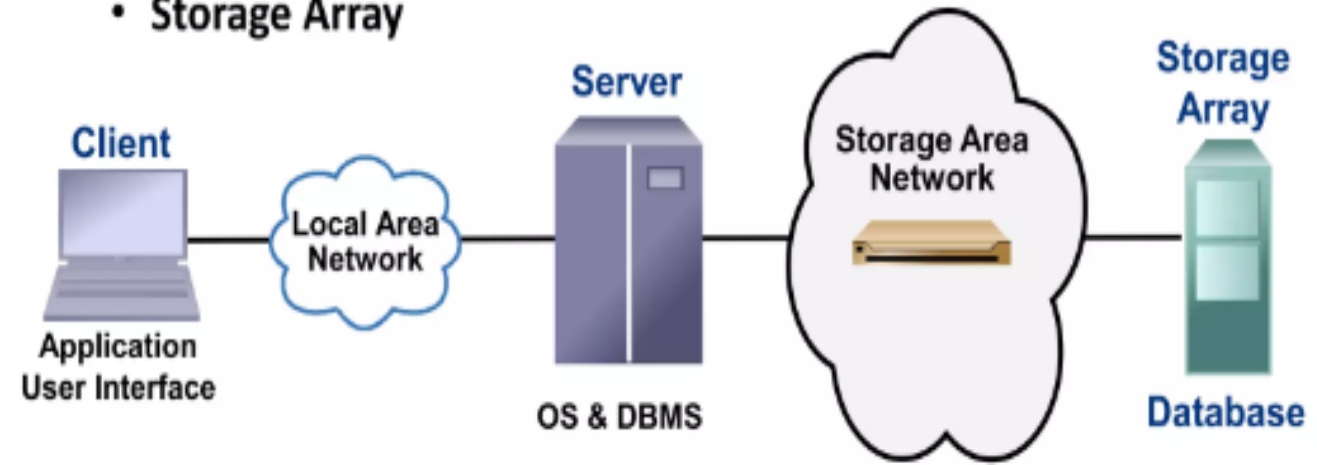
Order Entry System



- **Example: Order Entry System**

- **Consider an order processing system consisting of**

- Application [Order entry]
 - DBMS [Store customer and product information]
 - Server/OS [Application & Database programs are run]
 - Networks [Connectivity between Clients & Application/DB Server, Connectivity between Server & Storage system]
 - Storage Array



Data Center Infrastructure – Key Characteristics

- **Availability:** A data center should ensure the availability of information when required. Unavailability of information could cost millions of dollars per hour to businesses, such as financial services, telecommunications, and e-commerce.
- **Security:** Data centers must establish policies, procedures, and core element integration to prevent unauthorized access to information.
- **Scalability:** Business growth often requires deploying more servers, new applications, and additional databases. Data center resources should scale based on requirements, without interrupting business operations.
- **Performance:** All the elements of the data center should provide optimal performance based on the required service levels.

Data Center Infrastructure – Key Characteristics

- **Data integrity:** Data integrity refers to mechanisms, such as error correction codes or parity bits, which ensure that data is stored and retrieved exactly as it was received.
- **Capacity:** Data center operations require adequate resources to store and process large amounts of data, efficiently. When capacity requirements increase, the data center must provide additional capacity without interrupting availability or with minimal disruption. Capacity may be managed by reallocating the existing resources or by adding new resources.
- **Manageability:** A data center should provide easy and integrated management of all its elements. Manageability can be achieved through automation and reduction of human (manual) intervention in common tasks.

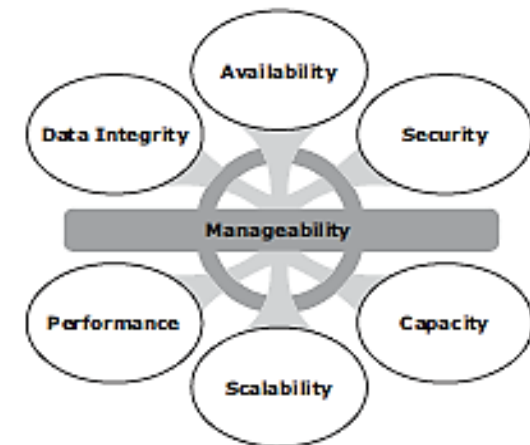


Figure : Key characteristics of a data center

Key Requirements for Data centre Elements

1. Availability
2. Location and facility
3. Uptime
4. Cooling
5. Security
6. Facility maintenance
7. Scalability
8. Change Management

Availability: All data centre elements should be designed to enable accessibility. The ability of users to access data when necessary can have a positive impact on a business.



Location and Facility: Location can have a considerable impact and could serve as an opportunity to enhance the company's ability to address network expectancy, disaster recovery, and data control. When determining the location of a data centre, it is also necessary to consider the geographical location with respect to the service provider.



Uptime: Organizations can lose a lot of money during the downtime due to the inaccessibility of necessary information and resources. The service provider's track record of uptime should be tested and it should be determined whether the provider has been able to maintain 100% availability for electrical and mechanical systems in the past.



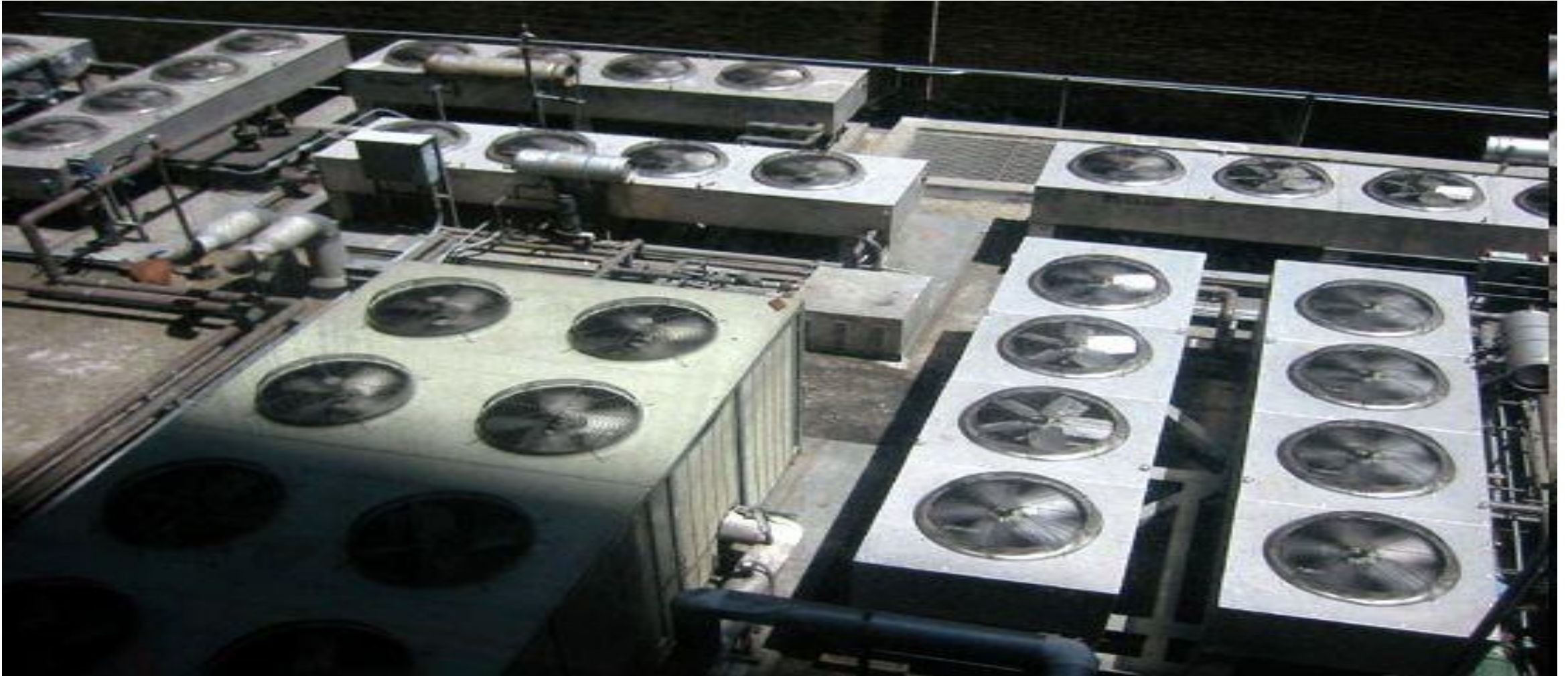
Powerful backup arrangement to provide 100% Uptime



Cooling: The establishment of a data centre involves a combination of utilities, generators, and a range of power supplies that emit excessive heat. It is essential to have a robust cooling infrastructure to ensure cost efficiencies and a viable environment to establish the data centre.



Cooling Systems



Security: The security of the data centre is equally important. It is necessary to establish policies and procedures, and ensure proper integration of the data centre core elements in order to prevent unauthorized access to information. To maximize protection, the facility should be fabricated with military-grade security measures.



Facility Maintenance: During the installation of a data centre, it is important to consider the maintenance and management aspects. The provider should have sufficient insight of the facility and its efficiencies to ensure that the entire system is being observed and any issues can be quickly addressed before they cause any outage.



Scalability: Data centre operations should be able to allocate additional on-demand storage or processing capabilities, without interrupting business operations. For the growth of a business, it is often necessary to install more servers, additional databases, and new applications. The storage solution should also be able to grow with the business.



Change Management: Proper guidelines for change management ensures that there is no alteration in the data centre that has not been planned, scheduled, discussed, and approved. It is also necessary to provide back out steps or a Plan B. Whether bringing new amendments to the system or making old practices obsolete, data centres must follow the change management process.



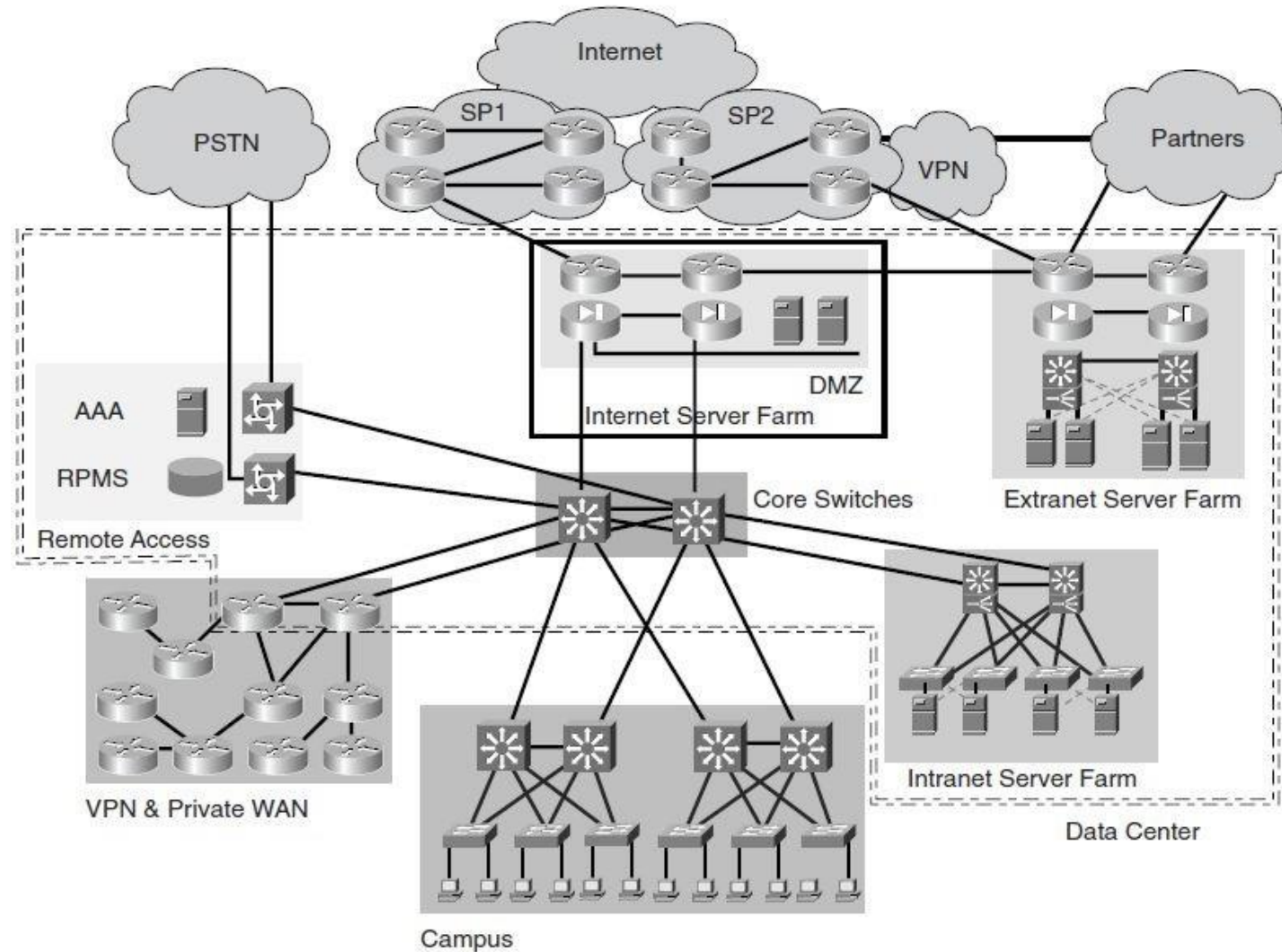
- **Example: Optimal Order Processing**
 - Application [Optimized for fast interaction with DBMS]
 - Database [The tables should be designed such that the number of R/W operations can be minimized]
 - Server [Contain sufficient CPU & memory resources to satisfy need of Application and DBMS]
 - Network [Provide fast communication between Client and Server, as well as Server and Storage Array]
 - Storage Array [Service R/W requests from Server for optimal performance]

The building blocks of a typical enterprise network includes the following:

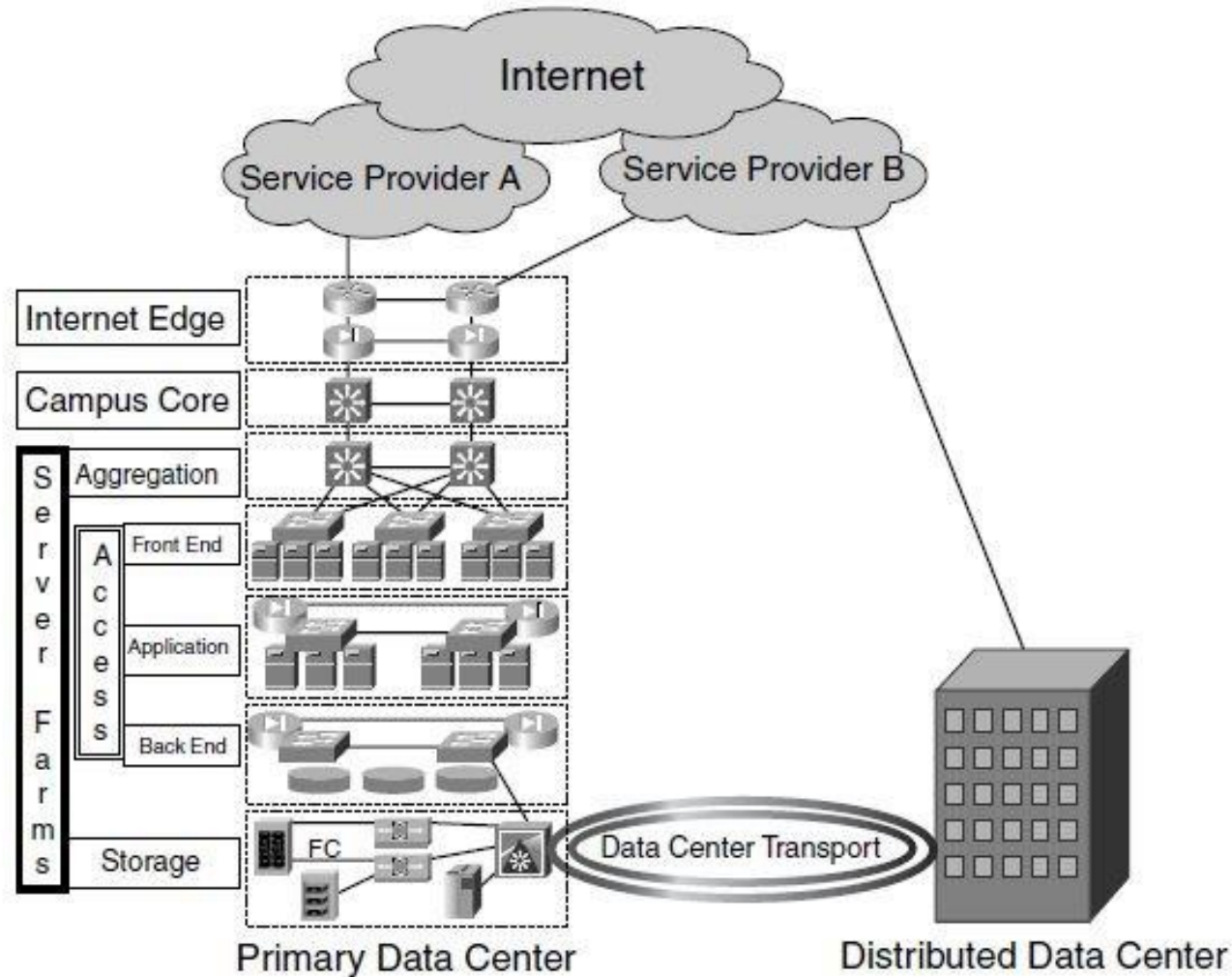
- Campus network
- Private WAN
- Remote access
- Internet server farm
- Extranet server farm
- Intranet server farm

Data Center – Enterprise

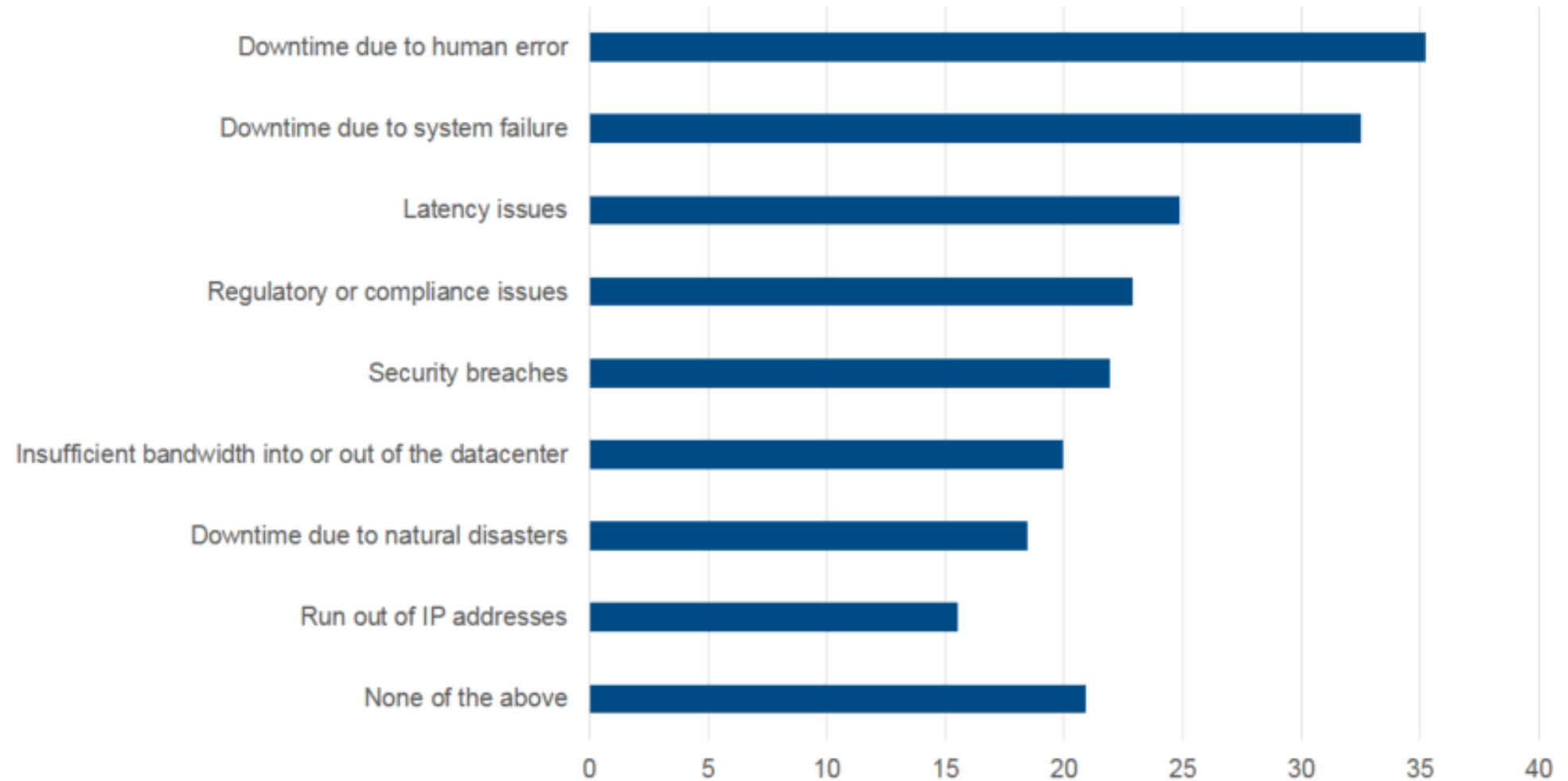
Data Centers in the Enterprise



Data Center – Enterprise Architecture



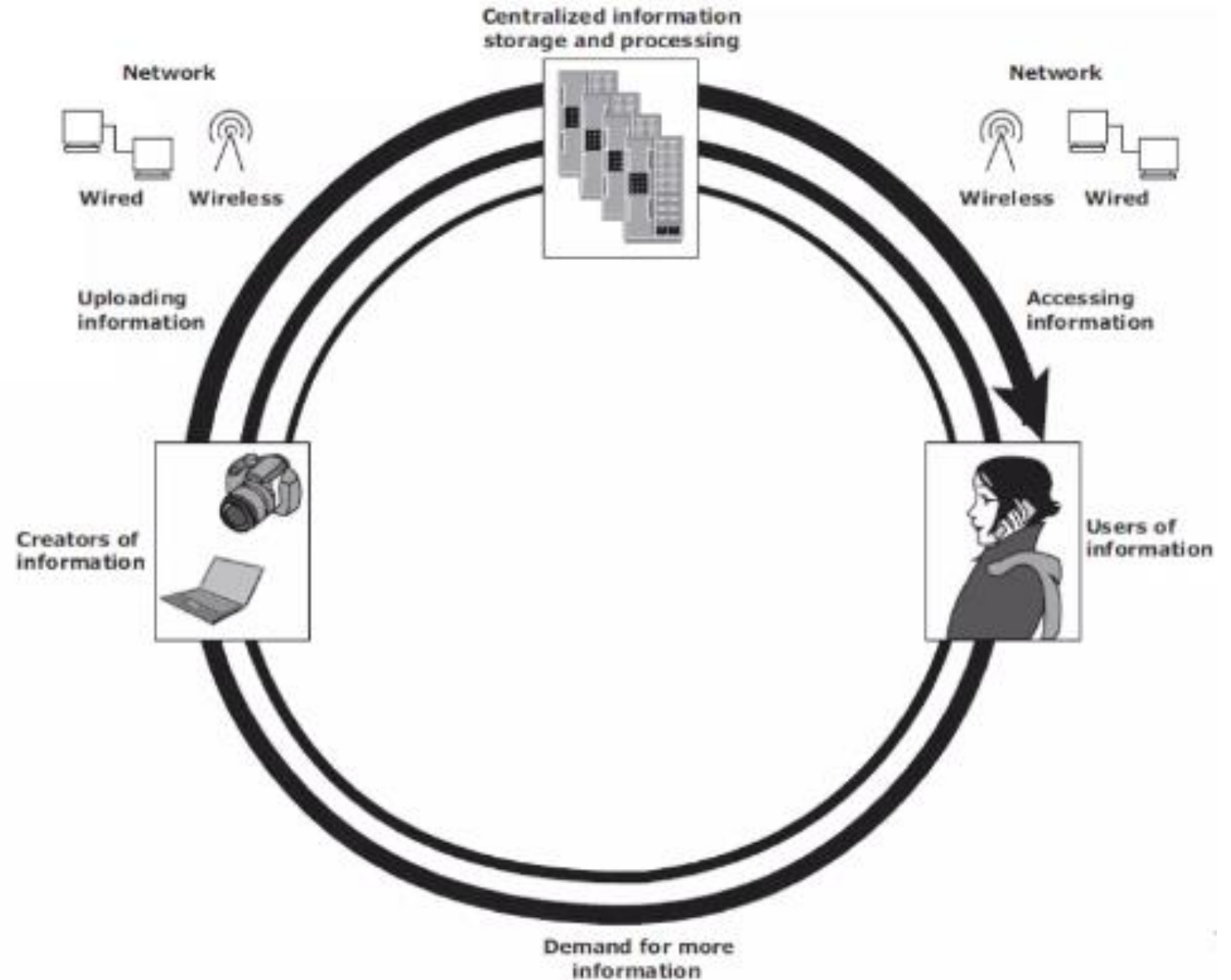
Data Center – Pitfalls



Data of Sunbird Company

Information Storage

Storing the process data or information and accessing it in useful ways



Information Storage

Here, list of some of the factors that have contributed to the growth of digital data

- **Increase in data-processing capabilities:** Modern computers provide a significant increase in processing and storage capabilities. This enables the conversion of various types of content and media from conventional forms to digital formats.
- **Lower cost of digital storage:** Technological advances and the decrease in the cost of storage devices have provided low-cost storage solutions. This cost benefit has increased the rate at which digital data is generated and stored.
- **Affordable and faster communication technology:** The rate of sharing digital data is now much faster than traditional approaches. A handwritten letter might take a week to reach its destination, whereas it typically takes only a few seconds for an e-mail message to reach its recipient.
- **Proliferation of applications and smart devices:** Smartphones, tablets, and newer digital devices, along with smart applications, have significantly contributed to the generation of digital content.

- Managing storage infrastructure is a key to ensure continuity of business
- Establishing management processes and implementing appropriate tools is essential to meeting service levels proactively
- Management activities include availability, capacity, performance, and security management
- Monitoring is the most important aspects that forms the basis for storage management
- Continuous monitoring enables availability and scalability by taking proactive measures

Big Data

- It includes unstructured and structured data
- The capacity and performance requirements of big data can be staggering, making advanced tactics critical for success.

Increased Flash Usage

- Flash technologies are rising in part because they are becoming accessible and in part because big data is making their performance capabilities necessary.
- Frequent access

Video Applications

- Many businesses are implementing video strategies as part of a larger goal to engage with employees and customers in more meaningful ways. While video offers incredible potential from an end-user perspective, it also creates nightmares for IT managers.

Legacy Systems

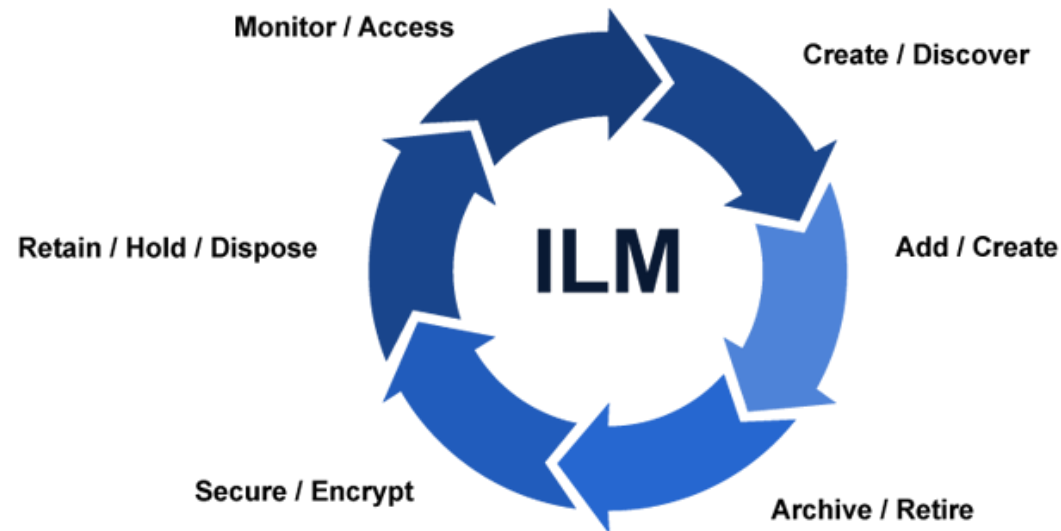
- The rapid rise in storage demands across the entire IT sector makes using legacy systems incredibly important for many organizations.
- Reliable storage systems that meet specific needs or handle different workflows particularly well can be incredibly valuable for IT teams

Data Protection

- While capacity and performance challenges are dominating the storage landscape, IT managers have a constant need to prevent data loss and protect data from theft.
- Third-party hardware maintenance plans can pay off in this area as well as vendors can help with processes like secure hard disk disposal and data recovery from damaged components.
- IT managers are facing staggering storage challenges, but the combination of advanced strategies and good partnerships can go a long way toward enabling IT leaders to keep pace with changing storage requirements.

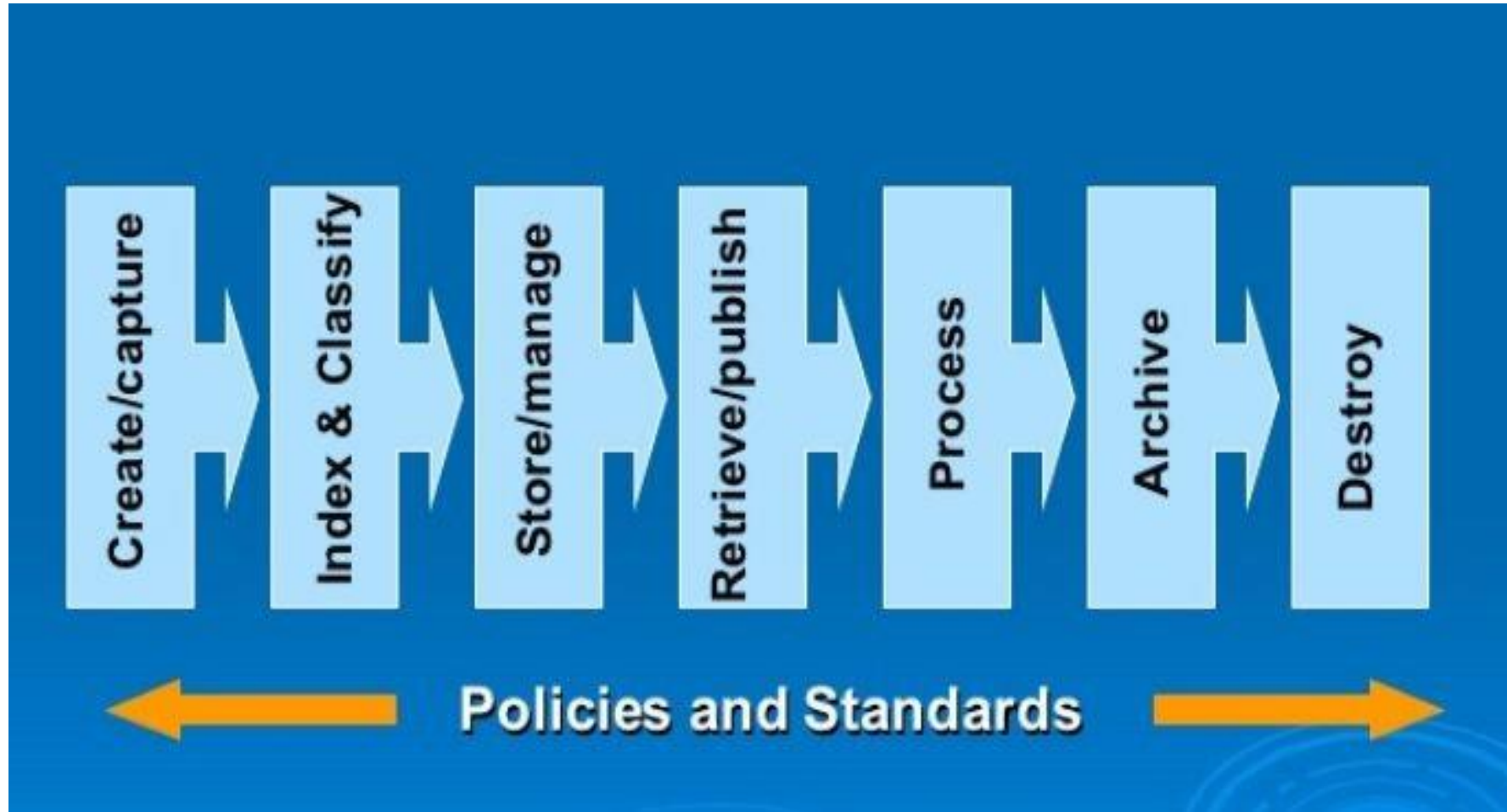
Information Life Cycle

- The information life cycle (ILM) is a way of managing data and storage that recognizes that the value of information changes over time.
- The goal of ILM is to oversee data from its creation through its retirement to optimize its usefulness, reduce costs, and minimize legal and compliance risks.



- In both traditional data center and virtualized environments, managing information can be expensive if not managed appropriately. Along with the tools, an effective management strategy is also required to manage information efficiently.
- This **strategy should address the following key challenges** that exist in today's data centers:
- **Exploding digital universe:** The rate of information growth is increasing exponentially. Creating copies of data to ensure high availability and repurposing has contributed to the multifold increase of information growth.
- **Increasing dependency on information:** The strategic use of information plays an important role in determining the success of a business and provides competitive advantages in the marketplace.
- **Changing value of information:** Information that is valuable today might become less important tomorrow. The value of information often changes over time.

- Stages of Information Life Cycle Management.



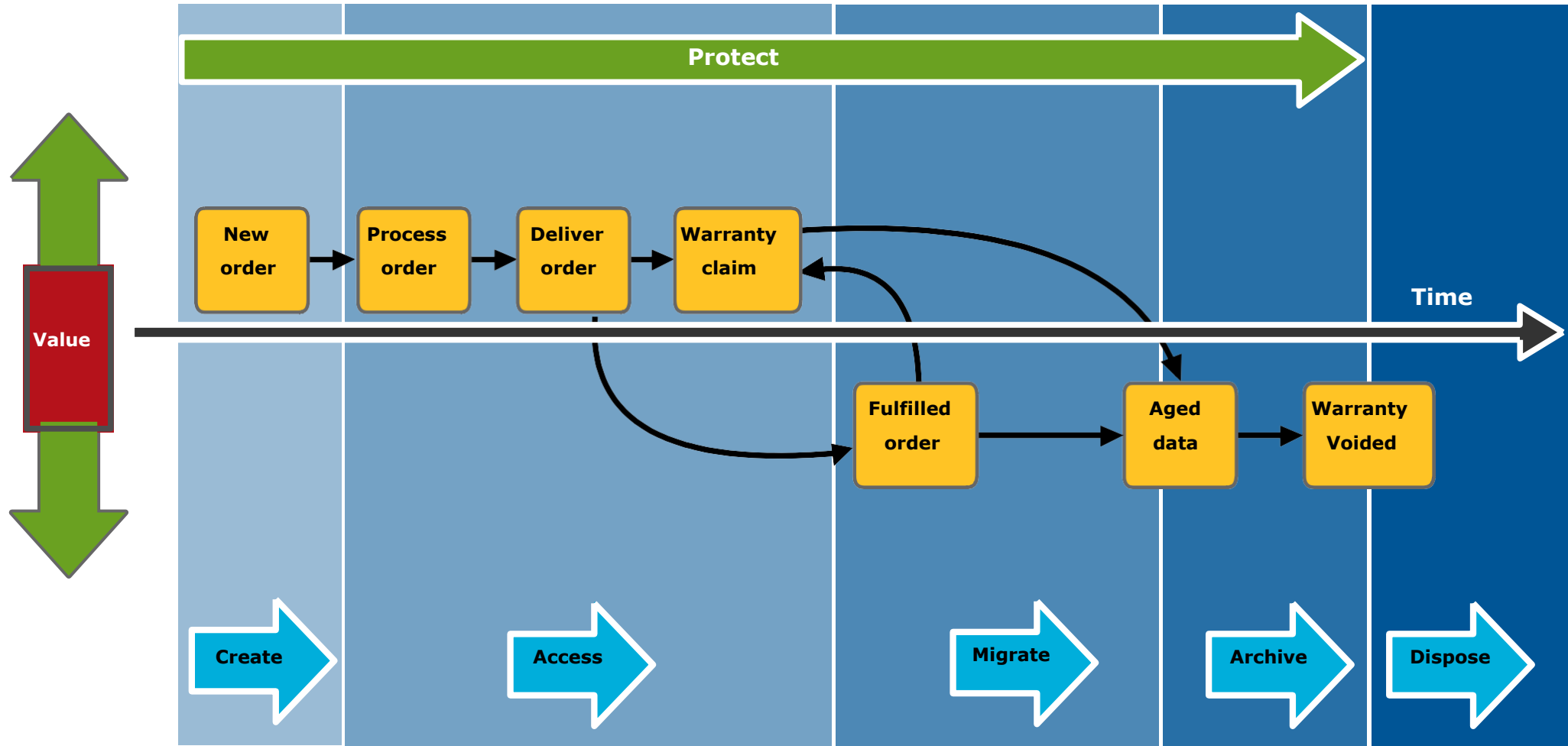


Figure: Changing value of sales order information

- ILM should possess the following characteristics to obtain and maintain an ILM strategy:



Business-centric



Policy-based



Optimised



Centrally-managed



Heterogeneous

How good ILM can be for your organisation?

- **Improved utilization**
 - Tiered storage platforms
- **Simplified management**
 - Processes, tools and automation
- **Simplified backup and recovery**
 - A wider range of options to balance the need for business continuity.
- **Maintaining compliance**
 - Knowledge of what data needs to be protected for what length of time.
- **Lower Total Cost of Ownership**
 - By aligning the infrastructure and management costs with information value.

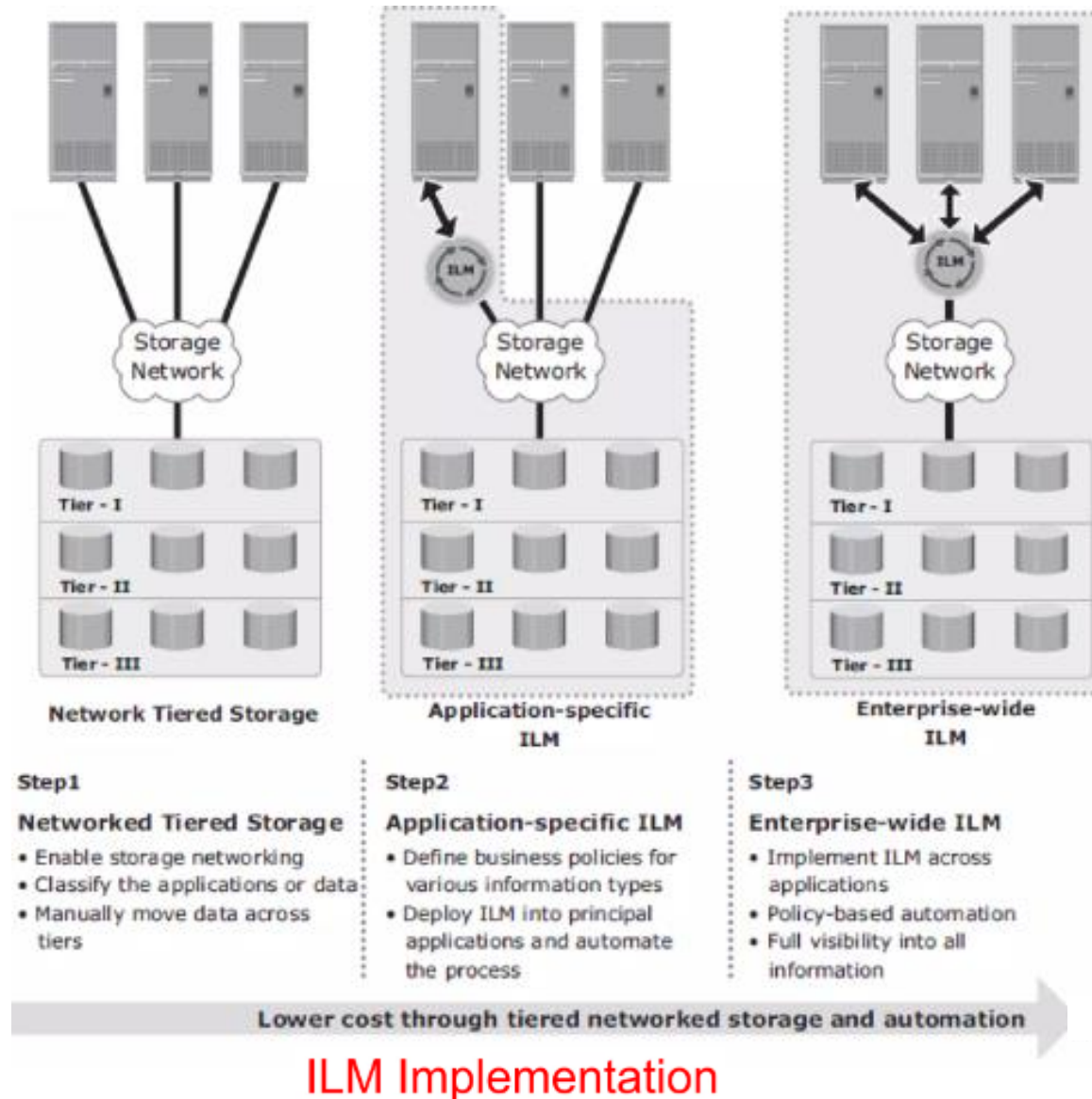
Information Life Cycle Management – Implementations

- **ILM Implementation**
 - **Classify data and applications**
 - Enable differentiated treatment of information on the basis of business rules and policies
 - **Implement policies**
 - Use of Information management tools from Data creation to Data disposal
 - E.g. EndNote, Mendeley, Papers, RefWorks, Zotero
 - **Managing the environment**
 - Use of integrated tools to reduce operational complexity
 - **Organizing storage resources in tiers**
 - Align the resources with data classes, and store information in the right type of infrastructure based on information's current value

Networked Tiered Storage: Classify the Apps, Moving across tiers

Application Specific: Business process and automate

Enterprise Wide: Policy based scheme



- **Improved utilization**
 - Tiered storage platforms increases visibility of all information
- **Simplified management**
 - Integrate process steps and interfaces with individual tools
 - Automation
- **A wider range of options for backup & recovery to balance the need for business continuity**
- **Maintaining Compliance**
 - Data needs to be protected for specified length of time
- **Lower Total Cost of Ownership (TCO)**
 - Align infrastructure & management costs with information value
 - Adv.: Resources are not wasted, Complexity is not introduced

Unit – 1 Summary

- Data is a collection of raw facts from which the required output is extracted.
- Information is processed data. The value of information changes over time.
- Storage technology has evolved through configurations such as RAID, DAS, SAN, NAS, and IP-SAN.
- Data Center and its Architecture discussed
- The change in value of information with the change in time is called information lifecycle.
- ILM is a comprehensive approach in managing the flow of data from its creation to its disposal.
- The key requirement for data centre, elements are availability, location and facility, uptime, cooling, security, facility maintenance, scalability, and change management.

Take Aways of this Unit I



- ✓ Information Storage
- ✓ Data – Types of Data –Information
- ✓ Storage, Evolution of Storage Technology and Architecture
- ✓ Data Centre Infrastructure
- ✓ Core elements
- ✓ Key Requirements for Data Centre Elements
- ✓ Managing Storage Infrastructure
- ✓ Key Challenges in Managing Information
- ✓ Information Lifecycle
- ✓ Information Lifecycle Management
- ✓ ILM Implementation -ILM Benefits

Resources

Topics	URL	Notes
Data centre Elements	https://www.techrepublic.com/blog/10-things/10-critical-elements-of-an-efficient-data-centre/	This link explains about typical elements that an efficient data centre is expected to have. The success of data centre depends on proper management of these elements
Data centre Basics	https://www.youtube.com/watch?v=XTwwKq7_HGY	This Video shows the components of Data Floating Data Center, Microsoft Undersea Data Center Explanation.

Resources

Topics	URL	Notes
Data centre Architecture	https://www.fiberoptics4sale.com/blogs/archive-posts/95041990-what-are-data-centres	This link explains about the detailed architecture of data centre, in particular enterprise data centre.
Data centre Strategy	http://focus.forsythe.com/articles/396/10-Steps-to-a-Successful-Data-centre-Strategy	This link explains the steps to a successful Data centre strategy.

Topics	URL	Notes
Data centre Deployment	https://www.youtube.com/watch?v=McVZ01kn-IQ	This video explains about real life data centre deployment and best practices.
Data centre Topology	https://www.youtube.com/watch?v=8M0XPOiKC7I	In this Video, you will learn about details of Data centre topology.

Resources

Topics	URL	Notes
Cover Story: Storage Management	http://www.networkmagazineindia.com/200309/coverstory01.shtml	This link explains a general overview of storage management, along with expert's opinion
Storage infrastructure management is still elusive	http://searchstorage.techtarget.com/opinion/Storage-infrastructure-management-is-still-elusive	You will learn Basic challenges associated with Storage and its management
Steps to ILM implementation	http://www.networkworld.com/article/2300188/data-centre/six-steps-to-ilm-implementation.html	This link explains about the six steps to successfully implement ILM
Information Lifecycle Management	http://searchstorage.techtarget.com/definition/information-life-cycle-management	This link explains about ILM approach in industry
Data centre Architecture	https://www.fiberoptics4sale.com/blogs/archive-posts/95041990-what-are-data-centres	This link explains about architecture of data centre , with emphasis to enterprise and discussing each component in details
Data centre Strategy	http://focus.forsythe.com/articles/396/10-Steps-to-a-Successful-Data-centre-Strategy	In this link, you will learn about the steps for adopting a successful Data centre Strategy

Resources

Topics	URL	Notes
Types of storage technologies	https://www.youtube.com/watch?v=bpUzGZLO948&list=PLWDUzz3hCDJJFiOO_zpMTbiHUNPDpU3xb	This link compares DAS, NAS and SAN technologies
Overview of data and information	https://www.youtube.com/watch?v=mUgEgkV16Bw	This link explains what is data, information and knowledge. A short description on how data differs from information.
Inside a Google Data centre	https://www.youtube.com/watch?v=XZmGGAbHqa0	This link takes us to a virtual tour inside a Google Data centre

Sample Test

Self Assessment Questions

1. Which one of the given options is true about data?
 - a) Raw fact
 - b) Processed fact
 - c) Processed information

Sample Test

Self Assessment Questions

1. Which one of the given options is true about data?
 - a) Raw fact
 - b) Processed fact
 - c) Processed information

Answer: a

Self Assessment Questions

2. Which of these is not an example of unstructured data?

- a) Images
- b) Videos
- c) Audio
- d) Spreadsheet

Self Assessment Questions

2. Which of these is not an example of unstructured data?

- a) Images
- b) Videos
- c) Audio
- d) Spreadsheet

Answer: d

Sample Test

Self Assessment Questions

3. Which one of the given options means On-command, on-demand world?

- a) Information needed when and where it is required.
- b) Data needed when and where it is required.
- c) Information not needed when and where it is required.
- d) Nothing needed when and where it is required.

Sample Test

Self Assessment Questions

3. Which one of the given options means On-command, on-demand world?

- a) Information needed when and where it is required.
- b) Data needed when and where it is required.
- c) Information not needed when and where it is required.
- d) Nothing needed when and where it is required.

Answer: a

Sample Test

Self Assessment Questions

4. Which one of the given options is a collection of raw facts from which conclusions may be drawn?

- a) Information
- b) Data
- c) Digital Value

Sample Test

Self Assessment Questions

4. Which one of the given options is a collection of raw facts from which conclusions may be drawn?

- a) Information
- b) Data
- c) Digital Value

Answer: b

Sample Test

Self Assessment Questions

5. Information created by both an individual and business. State whether True or False.

- a) True
- b) False

Self Assessment Questions

5. Information created by both an individual and business. State whether True or False.

- a) True
- b) False

Answer: a

References

- EMC Education Services, “Information Storage and Management: Storing, Managing, and Protecting Digital Information”, Wiley Publishing Inc., 2nd edition, 2012.
- <https://www.quora.com/p/20620/what-is-information-availability-and-information-1/>
- <https://www.edureka.co/blog/what-is-big-data/>
- <https://www.parkplacetechologies.com/blog/5-storage-challenges-facing-it-managers/>
- <https://www.cisco.com/c/en/us/solutions/data-center-virtualization/what-is-a-data-center.html>





THANK YOU